

SUPG & discontinuous Galerkin methods

Andrea Cangiani

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REFERENCES

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SUPG for advection–diffusion

Consider: find $u \in V = H_0^1(\Omega)$ such that

$$-\varepsilon \Delta u + \beta \cdot \nabla u + \mu u = f, \quad \varepsilon \text{ small.}$$

As in previous lecture, test with

$$v + \delta \beta \cdot \nabla v,$$

where δ is a parameter to be chosen later. This gives, formally,

$$\varepsilon(\nabla u, \nabla v) - \varepsilon(\delta \Delta u, \beta \cdot \nabla v) + (\beta \cdot \nabla u + \mu u, v + \delta \beta \cdot \nabla v) = (f, v + \delta \beta \cdot \nabla v).$$

showcasing the extra **streamline-diffusion (SD)** term.

Elementwise SD formulation

In fact, Δu_h is not globally well-defined.

So, we define the broken Laplacian Δ_h and set

$$a_h^{SD}(u, v) := \varepsilon(\nabla u, \nabla v) - \varepsilon(\delta \Delta_h u, \beta \cdot \nabla v) + (\beta \cdot \nabla u + \mu u, v + \delta \beta \cdot \nabla v),$$

$$\ell_h^{SD}(v) := (f, v + \delta \beta \cdot \nabla v).$$

Hence the **SD-FEM** is: find $u_h^{SD} \in V_h^k$ such that

$$a_h^{SD}(u_h^{SD}, v_h) = \ell_h^{SD}(v_h) \quad \forall v_h \in V_h^k.$$

Notes.

- ▶ The **SD term B** is added to enhance stability; the other terms (eg. the broken laplacian) are there for consistency.
- ▶ For linear elements, $\Delta_h u_h = 0$ on each element, so that terms disappears.

Stability and choice of δ_K

If

$$0 < \delta_K < \frac{1}{2} \min \left(\frac{\mu_0^2}{\|\mu\|_{L^\infty(K)}^2}, \frac{h_K^2}{\varepsilon C_{inv}^2} \right),$$

where C_{inv} is the constant in the local inverse inequality

$$\|\Delta v_h\|_{L^2(K)} \leq C_{inv} h_K^{-1} |v_h|_{H^1(K)}, \quad \forall K \in \mathcal{T}_h, \quad \forall v_h \in V_h^k,$$

then

$$a_h^{SD}(v_h, v_h) \geq \frac{1}{2} \|v_h\|_{SD}^2,$$

with

$$\|v_h\|_{SD}^2 := \varepsilon |v_h|_{H^1(\Omega)}^2 + \sum_K \delta_K \|\beta \cdot \nabla v_h\|_{L^2(K)}^2 + \mu_0 \|v_h\|_{L^2(\Omega)}^2.$$

A priori error estimate

Define the mesh Péclet number

$$Pe_K = \frac{\|\beta\|_{L^\infty(K)} h_K}{2\varepsilon}.$$

Choose

$$\delta_K = \begin{cases} \delta_0 h_K, & Pe_K > 1, \\ \delta_1 h_K^2 / \varepsilon, & Pe_K \leq 1. \end{cases}$$

Then

$$\|u - u_h^{SD}\|_{SD} \leq C (\varepsilon^{1/2} + h^{1/2}) h^k |u|_{H^{k+1}(\Omega)}.$$

DG for transport [Reed–Hill, LANL Research report, 1973]

We start directly from the **divergence-form transport problem** used in the deal.II tutorial:

$$\nabla \cdot (\beta u) = 0 \quad \text{in } \Omega, \quad u = g \quad \text{on } \partial\Omega^-.$$

Here

$$\partial\Omega^- := \{x \in \partial\Omega : \beta(x) \cdot n(x) < 0\}, \quad \partial\Omega^+ := \partial\Omega \setminus \partial\Omega^-.$$

Assumptions. $\Omega \subset \mathbb{R}^d$ polyhedral, $\beta \in [L^\infty(\Omega)]^d$, $\nabla \cdot \beta \in L^\infty(\Omega)$ with

$$-\frac{1}{2} \nabla \cdot \beta \geq \mu_0 > 0.$$

We shall work with **graph space**

$$W := \{w \in L^2(\Omega) : \beta \cdot \nabla w \in L^2(\Omega)\}.$$

Hilbert with $(v, w)_W := (v, w) + (\beta \cdot \nabla v, \beta \cdot \nabla w)$ inducing norm

$$\|w\|_W^2 = \|w\|_{L^2(\Omega)}^2 + \|\beta \cdot \nabla w\|_{L^2(\Omega)}^2.$$

A weak formulation

Test with $v \in H^1(\Omega)$ and integrating by parts:

$$\int_{\Omega} \nabla \cdot (\beta u) v = - \int_{\Omega} u \beta \cdot \nabla v + \int_{\partial\Omega} (\beta \cdot n) u v = 0.$$

Imposing weakly the boundary condition,

$$- \int_{\Omega} u \beta \cdot \nabla v + \int_{\partial\Omega^+} (\beta \cdot n) u v = \int_{\partial\Omega^-} (\beta \cdot n) g v.$$

Towards a dG formulation: Using a finite element space made of piecewise continuous functions with respect to the mesh, we **cannot integrate by parts globally...**

Elementwise integration by parts

Assume Ω polygonal / polyhedral and let $\{\mathcal{T}_h\}_h$ be a family of meshes.

For each element $K \in \mathcal{T}_h$, test with v and integrate by parts:

$$\int_K \nabla \cdot (\beta u) v = - \int_K u \beta \cdot \nabla v + \int_{\partial K} (\beta \cdot n_K) u v.$$

Hence the local equation for the exact solution is

$$- \int_K u \beta \cdot \nabla v + \int_{\partial K} (\beta \cdot n_K) u v = 0.$$

Information enters K through the inflow part of its boundary:

$$\partial K^- := \{x \in \partial K : \beta(x) \cdot n_K(x) < 0\}, \quad \partial K^+ := \{x \in \partial K : \beta(x) \cdot n_K(x) > 0\}.$$

so we rewrite the local equation as

$$- \int_K u \beta \cdot \nabla v + \int_{\partial K^+} (\beta \cdot n_K) u v + \int_{\partial K^-} (\beta \cdot n_K) u v = 0.$$

Upwind traces

For a discontinuous approximation we must specify a **numerical trace** on every face. Pick **upwind choice**:

- ▶ if $F \subset \partial\Omega^-$, then $u^{up} = g$, (**weak imposition** of boundary cond.¹)
- ▶ if $F \subset \partial\Omega^+$, then $u^{up} = u|_K$,
- ▶ if F is interior, then u^{up} is the **trace coming from the neighbouring element**.

If an interior face F is shared by K^+ and K^- , choose the normal n^+ so that

$$\beta \cdot n^+ > 0.$$

Then define

$$[v] := v^+ - v^-.$$

The corresponding face term is written as

$$\int_F \beta \cdot n^+ u^{up} [v].$$

¹First proposed and known as method of [Nitsche, Hamburg rep., 1971].

DG space and global formulation

Define the **discontinuous finite element** space

$$V_h := \{v_h \in L^2(\Omega) : v_h|_K \in \mathbb{P}_k(K) \ \forall K \in \mathcal{T}_h\}.$$

Summing the local relations over all cells gives:

Discontinuous Galerkin (dG) formulation: find $u_h \in V_h$ such that

$$\begin{aligned} - \sum_{K \in \mathcal{T}_h} \int_K u_h \beta \cdot \nabla v_h + \sum_{F \in \mathcal{F}_h^i} \int_F \beta \cdot n^+ u^{up}[v_h] + \sum_{F \in \mathcal{F}_h^+} \int_F \beta \cdot n u^{up} v_h \\ = - \sum_{F \in \mathcal{F}_h^-} \int_F \beta \cdot n g v_h \end{aligned}$$

for all $v_h \in V_h$.

Here $\mathcal{F}_h^i$ denotes the set of interior faces, while

$$\mathcal{F}_h^- = \{F \subset \partial\Omega^-\}, \quad \mathcal{F}_h^+ = \{F \subset \partial\Omega^+\}.$$

Remarks

1. **Sparsity pattern.** Communication between degrees of freedom only through mesh faces.
2. **Implementation.** Once the inflow value on ∂K^- is known, each elemental problem can be solved independently. In favorable cases, one can march from the inflow boundary without building the full global matrix.
3. **Galerkin orthogonality.** By construction the method is consistent, hence

$$B(u - u_h, v_h) = 0 \quad \forall v_h \in V_h.$$

4. **Stability mechanism.** If β has constant components, then

$$v_h + \delta \beta \cdot \nabla v_h \in V_h \quad \forall \delta \geq 0,$$

thanks to the richness of the DG space. Combined with weak imposition of the inflow data, this yields stability in an L^2 -type graph norm (see eg. Ern-Guermond).

Error bound

Theorem (error bound)

Assume that $u \in H^{p+1}(\Omega)$. Then,

$$\|u - u_h\|_{L^2(\Omega)} + \left(\sum_{F \in \mathcal{F}_h^i} \left\| \frac{|\beta \cdot n|^{1/2}}{2} [u - u_h] \right\|_{L^2(F)}^2 \right)^{1/2} \leq Ch^{p+1/2} |u|_{H^{p+1}(\Omega)}.$$

DG for Poisson: starting point

We now switch to Poisson with non-homogeneous Dirichlet boundary conditions:

$$-\Delta u = f \quad \text{in } \Omega, \quad u = g \quad \text{on } \partial\Omega.$$

Assume,

$$u \in V_* := H^1(\Omega) \cap W^{2,1}(\Omega),$$

so that the normal trace $\nabla u \cdot n$ is well defined on element boundaries.

Testing elementwise with $v \in V_h$ and summing over the mesh gives

$$\sum_{K \in \mathcal{T}_h} \int_K \nabla u \cdot \nabla v - \sum_{K \in \mathcal{T}_h} \int_{\partial K} (\nabla u \cdot n) v = \int_{\Omega} f v.$$

Define

$$a_h(u, v) := \sum_{K \in \mathcal{T}_h} \int_K \nabla u \cdot \nabla v.$$

Jumps, averages, and consistency terms

On an interior face F shared by two elements, define

$$\{\{w\}\} := \frac{1}{2}(w^+ + w^-), \quad \llbracket w \rrbracket := w^+ n^+ + w^- n^-.$$

For vector fields,

$$\{\{\tau\}\} := \frac{1}{2}(\tau^+ + \tau^-), \quad \llbracket \tau \rrbracket := \tau^+ \cdot n^+ + \tau^- \cdot n^-.$$

Extend the same notation to boundary faces by taking the exterior value equal to 0.

Then the exact solution satisfies the consistent form

$$a_h(u, v) - \sum_{F \in \mathcal{F}_h} \int_F \{\{\nabla u\}\} \cdot \llbracket v \rrbracket ds - \sum_{F \in \mathcal{F}_h^b} \int_F (u - g) \nabla v \cdot n ds = \int_{\Omega} f v.$$

To recover symmetry, add the zero term

$$- \sum_{F \in \mathcal{F}_h^i} \int_F \llbracket u \rrbracket \cdot \{\{\nabla v\}\} ds - \sum_{F \in \mathcal{F}_h^b} \int_F (u - g) \nabla v \cdot n ds.$$

SIPDG formulation

This leads to the symmetric interior penalty DG method. Define

$$B_h^S(w, v) := a_h(w, v) - \sum_{F \in \mathcal{F}_h} \int_F \left(\{\nabla w\} \cdot \llbracket v \rrbracket + \{\nabla v\} \cdot \llbracket w \rrbracket \right) ds,$$

and add the **penalty term**

$$b_h(w, v) := B_h^S(w, v) + \sum_{F \in \mathcal{F}_h} \int_F \sigma \llbracket w \rrbracket \cdot \llbracket v \rrbracket ds.$$

The right-hand side is

$$\ell_h(v) := \int_{\Omega} f v - \sum_{F \in \mathcal{F}_h^b} \int_F g \nabla v \cdot n ds + \sum_{F \in \mathcal{F}_h^b} \int_F \sigma g v ds.$$

The SIP-DG method is: find $u_h \in V_h$ such that

$$b_h(u_h, v_h) = \ell_h(v_h) \quad \forall v_h \in V_h.$$

The penalty is chosen as $\sigma|_F = \sigma_0 p^2 \frac{|F|}{|K|}$, for $\sigma_0 > 0$ big enough, which yields a stable and optimal method.